AI-Assisted Federal Survey Harmonization

Research Fact Sheet

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What's the problem?

This research examines **47 Census Bureau demographic survey instruments** containing approximately **7,000 questions**, each designed for a specific analytical purpose. No complete map exists showing where these surveys overlap, where their questions are compatible, and where the differences matter. Building that map manually represents months of effort for a skilled research team. This research accomplished that work in weeks using AI-assisted methods, at a total API cost under $100.

Why should leadership care?

Response rates are declining across surveys. Every year, it costs more to collect less. Before looking to alternative data sources, we should be in-sourcing: getting more value from data we already collect.

The Census Bureau already *has* enormous data that could answer questions no single survey can. Cross-survey integration — using overlapping questions as bridge variables — increases the explanatory power of existing collections without adding respondent burden. Where questions turn out to be interchangeable, the same analysis identifies consolidation opportunities and encourages language standardization across instruments, improving data quality and fit-for-purpose alignment.

How is this different?

Traditional survey harmonization is manual, slow, and limited to 2–3 surveys at a time. This research uses **multiple AI models from different vendors independently rating question pairs**, avoiding single-model bias. The approach holds all 47 surveys in view simultaneously, finding connections no individual expert could surface — not because experts lack skill, but because no one holds 7,000 questions in working memory at once.

Does it work?

Three independent AI models from competing vendors reach near-perfect agreement on harmonization judgments, confirming the task is well-defined and results are not driven by any single model's bias. Harmonization rates are stable across different survey pairs (CPS and FoodAPS each scored against ACS), suggesting a general property of the data rather than a survey-specific artifact. The framework passes basic sanity checks: demographics like race and age score as near-100% consolidable, while genuinely different constructs are correctly flagged with specific barrier diagnoses. The full pipeline, data, and code are public and reproducible.

Key findings

~45% of questions across two major survey pairs (CPS and FoodAPS vs. ACS) have at least one viable harmonization path:

- **Tier 1 – Direct recode** (~16%): Interchangeable with simple recoding. High-confidence bridge variables.
- **Tier 2 – Statistical adjustment** (~29%): Linkable with documented transformations. Usable bridges with known limitations.
- **Tier 3 – Not linkable** (~55%): Different constructs. The framework identifies these and explains why.

These results give survey managers a scored inventory of where cross-survey integration adds value and where instruments can be aligned for better return on collection investment.

What’s at stake?

Without this work, 47 surveys continue as isolated silos. Declining response rates erode quality with no compensating strategy. Connecting existing surveys increases the overall value of current collections, helps each survey collect the data it’s best positioned to gather, and builds resilience as the data environment shifts. The methodology extends naturally to surveys across the broader federal statistical system.

What’s next?

Stage 4 moves from pairwise comparisons to the full multi-survey network, discovering multi-hop integration paths (Survey A to B to C) across all 47 Census demographic surveys.
